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GitHub Repo Link: https://github.com/afc9jk/Hackathon

Video Link: https://youtu.be/TvTa_QeAYYM

Project Overview

Our group built a nine-metric safety and quality scorecard for evaluating the gemma-4-E2B-it large language model as a personal-finance assistant, with a specific focus on credit disputes, identity theft, and financial fraud. This is a vital domain to evaluate because potential advice-seekers will frequently be in crisis, professional help for their time-sensitive issue is often expensive or inaccessible, and mistakes they are advised to make are permanently damaging. We use judges to evaluate the gemma-4 model responses for 40 queries against a baseline optimal LLM response. We also test cross-lingual safety distribution shift for Spanish and Swahili. Ultimately, we produce contextualized scores while being explicit about the limitations of our data, metrics, and process.

Evaluation Methodology

Sampling Strategy

Rather than running inference and multi-judge scoring across all 7,036 rows in our dataset, we drew a small, stratified sample across the dataset's eight category labels so that small categories (e.g., Estate Planning & Legacy, 1.0% of the dataset) were not drowned out by large ones (e.g., Debt Management & Credit, 28.3%). This kept our compute budget within the six-hour session and single-GPU allocation while still giving every category enough representation to compute a meaningful per-category score. Each data row represents a very complex query and response, so we are not concerned that we only used a small portion of this very comprehensive dataset.

Inference Pipeline

We adapted the provided inference_boilerplate.py to run gemma-4-E2B-it over our sampled queries using its chat template (system and user turns), generating one response per query. Inference was run in parallel on the Rivanna GPU.

Judge Panel

Judge 1: ShieldGemma-2b: Google's ShieldGemma-2b, used largely as-is via the provided llm_judge_boilerplate.py. This model scores P(violation) via softmax over Yes/No logits for our 8 metrics we are measuring.

We wanted to run multiple judges and average their scores, but the GPU time was a huge constraint, so we stuck to 1. Even though we couldn't use two, we still standardized our judge's output format (a constrained probability score per rubric dimension) so that the scores would theoretically be comparable across judges if we had the time and resources to do multiple.

Project Discussion: Use Case and Motivation

Why We Chose Our Finance Use Case

Our personal finance fraud choice spans eight categories: debt management and credit, retirement planning, investing and wealth building, tax planning, budgeting, insurance, savings, and estate planning. We chose this domain for three reasons as listed below:

First, finance is a domain that almost everyone interacts with; through their mortgages, credit cards, and tax preparation. But not many people are formally trained in it. Because of this, normal people turn to free, instantly accessible tools like chatbots, because CPAs, financial advisors, and attorneys are expensive and hard to access. This unfortunately means that an LLM's mistakes are impressed upon the population least equipped to identify and catch them.

Second, the dataset is incredibly rich and complex. It is real Reddit questions annotated with inferred psychological intent and emotional state, which let us evaluate not just factual correctness but whether a model responds appropriately to someone who is frightened, angry, or confused, which most generic safety benchmarks ignore.

Third, credit fraud and identity theft cause both financial harm and privacy harm: victims in this dataset are often asked to share account details, Social Security-adjacent information, and other sensitive data in the course of describing their problem, so this use case let us evaluate both financial-advice safety and PII handling in a single, coherent scorecard.

Why This Is a High-Stakes Area

Financial harm is often irreversible: unlike a wrong movie recommendation, bad financial advice compounds. Advising someone to raid a retirement account, miss a debt-dispute deadline, or ignore a fraudulent tax filing can cause damage that persists for years and is difficult or impossible to reverse.

Users are frequently already in crisis: several categories in our dataset (identity theft, collections disputes, tax fraud) involve users who are already victims of a crime. A model that responds insensitively, gives incorrect procedural guidance (e.g., wrong agency, wrong deadline, wrong legal right), or fails to recognize signs of fraud can make an already difficult situation meaningfully worse.

Regulatory and legal exposure: unlike many chatbot use cases, financial advice is subject to real regulatory frameworks (FCRA, IRS procedures, state-specific consumer protection law). Getting these details wrong is not merely unhelpful; it can lead a user to take an action with real legal or financial consequences.

High variance in what "correct" advice even means: the same query can call for very different advice depending on income, debt load, age, risk tolerance, and jurisdiction. A model that gives generic, one-size-fits-all financial advice is not just unhelpful; it can be actively harmful if the user's actual situation diverges from the implicit assumptions baked into that advice.

Overall, we treated this hackathon not as a generic chatbot-quality exercise but as an exercise in identifying where a general-purpose LLM's architecture and training genuinely limits its ability to act safely in a high-stakes financial domain.

Dataset Overview

About the Dataset

We use Akhil-Theerthala/PersonalFinance_v2, a Hugging Face dataset of 7,036 real personal-finance questions sourced from Reddit's r/personalfinance (restricted to posts predating Reddit's mid-2023 API/ToS changes for clean, ethical sourcing). Each row contains four fields: the original user query, a structured chain-of-thought reasoning trace, a final polished response, and a category label. Responses were produced through a multi-model pipeline: candidate answers were generated with retrieved context and inferred user sentiment/intent, then scored by an LLM jury (GPT-4o-mini, Deepseek-v3-0324, and Gemini-2.0-Flash), with the highest-rated candidate kept as the final response. It is released under the Apache 2.0 license.

Summary Statistics

The dataset has no missing values and no duplicate rows across its 7,036 entries. Category representation is heavily imbalanced, and the three text fields differ substantially in length, as shown below.

Category	Rows	% of Total	Notes
Debt Management & Credit	1,989	28.3%	Largest category; central to our credit/fraud use case.
Retirement Planning	1,169	16.6%	
Investing & Wealth Building	1,156	16.4%	
Tax Planning & Optimization	1,019	14.5%	
Budgeting & Cash Flow Mgmt.	986	14.0%	
Insurance & Risk Management	390	5.5%	
Savings & Emergency Funds	259	3.7%	
Estate Planning & Legacy	68	1.0%	Smallest category; limited statistical power.

Field	Mean (words)	Median	Min	Max
query (question)	165.4	131	14	2,003
chain_of_thought	1,083.7	1,074	585	1,763
response (answer)	340.8	341	56	682

The chain_of_thought field is consistently long and dense, while query length is much more variable, which reflects real-world Reddit post lengths, which can range from one-line questions to multi-paragraph accounts of a financial or fraud situation. Response length is consistent, usually around 300–400 words.

Metrics

The hackathon info sheet supplied a broad menu of potential safety dimensions across six categories. Below, we map every suggested metric to our final nine metrics. We wanted to explain our decision-making process on why we either adopted it directly, adapted it, or excluded it. We want our scorecard to be completely transparent.

Category	Suggested Metric	Decision	Rationale
Truthfulness & grounding	Faithfulness / hallucination rate	Adopted	Directly became Metric 2 (Hallucination / Unsupported Claim Rate), reframed around tax, legal, rate, and deadline facts specific to our domain.
Truthfulness & grounding	Citation accuracy / source attribution	Excluded	The dataset's responses are not built around citing external sources, so there is no source-attribution behavior to score; not applicable to this data.
Truthfulness & grounding	Factual consistency across a conversation	Excluded	Our evaluation is single-turn (one query, one response), so there is no multi-turn conversation in which the model could contradict itself; flagged as future work if multi-turn data is added.
Truthfulness & grounding	Calibration (confidence vs. accuracy)	Excluded	Would require the model to output explicit confidence scores, which our prompting setup does not elicit; out of scope given time constraints.
Robustness & security	Robustness to rephrasing/noise	Adopted	Directly became Metric 6 (Robustness to Rephrasing).
Robustness & security	Attack success rate / jailbreak resistance	Excluded	Our use case is everyday financial-advice safety, not adversarial red-teaming; a proper jailbreak evaluation needs a dedicated attack dataset outside PersonalFinance_v2's scope.
Fairness & bias	Bias / performance disparity across demographic, dialect, or political axes	Excluded	The dataset does not label user demographics, so we cannot reliably measure performance disparity across groups; noted as a real gap in our Limitations section rather than papered over.
Fairness & bias	Stereotype reinforcement rate	Excluded	No demographic ground truth to check against.
Privacy & data	PII leakage rate	Adapted	Folded into Metric 3 (Harmful Recommendation Rate) and Metric 9 (Personalization), since our domain's queries often contain real sensitive details (account info, SSN-adjacent data); not scored as a fully independent quantitative metric due to time constraints.
Privacy & data	Training-data memorization / regurgitation	Excluded	Requires a specialized membership-inference or extraction probe set; out of scope for a finance-Q&A evaluation.
Privacy & data	Context-confidentiality (cross-session leakage)	Excluded	Not applicable. Our test harness evaluates independent single-turn queries with no shared session state between users.
Behavioral / alignment	Refusal behavior (over- and under-refusal)	Adapted	Became Metric 8 (Uncertainty / Referral Behavior), a domain-specific version: the safe behavior in finance is rarely a flat refusal, but a calibrated referral to a professional or a request for missing details.
Behavioral / alignment	Sycophancy	Excluded	Not scored as a separate metric; sycophantic agreement with a bad financial plan should already surface as lower Advice Accuracy, so we treat it as implicitly covered rather than double-counted.
Behavioral / alignment	Toxicity	Excluded	Since our domain primarily focuses on advice, pertain more to reasoning quality and the

Category	Suggested Metric	Decision	Rationale
			harmfulness of a recommendation as opposed to toxicity in answers.
Behavioral / alignment	Instruction-following fidelity	Adapted	Folded into Metric 4 (Completeness / Actionability), which checks whether the response actually addresses what was asked.
Behavioral / alignment	Steerability (system-prompt constraints)	Excluded	Would need a separate test harness varying system prompts (e.g., “never give exact dosages” equivalents for finance); flagged as future work.
Reasoning quality	Explainability	Adopted	Became part of Metric 5 (Financial Reasoning Quality).
Reasoning quality	Plausibility	Adopted	Also folded into Metric 5.
Reasoning quality	Reasoning faithfulness (stated CoT matches actual computation)	Explicitly excluded	We cannot verify a model's internal computation, only its stated output, so Metric 5 deliberately scores observable reasoning quality only. This exclusion is intentional and documented, not an oversight.
Practical / deployment	Latency & cost	Excluded (discussed qualitatively)	Not scored numerically, but our GPU/judge-sequencing constraints (see Evaluation Methodology and Limitations) amount to a practical latency/cost discussion for this deployment context.
Practical / deployment	Failure-mode graceful degradation / abstention rate	Adapted	Connected to Metric 8 (Uncertainty / Referral Behavior). Knowing when to defer is our version of graceful degradation.
Practical / deployment	Coverage / answer rate	Adapted	Connected to Metrics 4 and 8 together: how often the model actually attempts a complete answer vs. appropriately defers.

Two categories of the original menu: Fairness & Bias and several Privacy/Deployment sub-metrics, are the most significant gaps we are knowingly leaving open, purely because our dataset lacks the demographic labels and multi-turn/session structure needed to measure them properly. We would rather state this clearly than force a metric onto data that cannot support it.

Detailed Metric Descriptions

Below we detail each of the nine metrics in our scorecard: what it captures, why it belongs in a personal-finance / credit-fraud scorecard specifically, how we measured it, and its known limitations.

1. Advice Accuracy / Financial Correctness

What it measures: Whether the model's advice is financially sound: is it consistent with basic financial rules, correct given the specific facts in the query, and not contradicted by well-established personal-finance principles.

Why it belongs in this scorecard: This is our core metric. Everything else in the scorecard (safety, tone, completeness) is secondary if the underlying advice is simply wrong. An eloquent, well-caveated, but factually incorrect answer can still cause real financial harm.

How it is measured: LLM-as-judge, prompted with the query, the model's response, and the dataset's reference response as context, scored against a rubric for financial soundness and internal consistency. Judged per-category so that category-specific correctness (e.g., FCRA rules for credit disputes vs. contribution limits for retirement accounts) is captured rather than averaged away.

Limitations: The reference response is itself LLM-generated (see Limitations, item 1), so "correctness" is bounded by both the judge model's own financial knowledge and the reference's accuracy. This metric cannot substitute for review by an actual financial professional.

2. Hallucination / Unsupported Claim Rate

What it measures: The rate at which the model states specific, checkable facts, like tax rules, laws, interest rates, regulations, account limits, deadlines, that end up being fabricated, outdated, or unsupported, framed specifically as "unsupported or fabricated financial claims" rather than generic hallucination.

Why it belongs in this scorecard: Financial advice is dense with specific, checkable numbers and rules (contribution limits, filing deadlines, statute-of-limitations windows for debt). A single fabricated deadline or dollar figure can lead a user to miss a real legal window or make an incorrect financial decision.

How it is measured: LLM-as-judge extracts each discrete factual claim from the response, classifies it as supported / unsupported / contradicted relative to the reference response and known domain facts, and reports the proportion of unsupported or fabricated claims per response.

Limitations: Judges can only flag claims against their own training knowledge or the (possibly outdated, see Limitations item 5) reference response. A claim that is wrong but plausible-sounding and unchallenged by the reference may not be caught.

3. Harmful Recommendation Rate

What it measures: The rate at which the model actively recommends an action that is financially or legally dangerous. For example, casually recommending an early retirement-account withdrawal, telling a user to ignore taxes or a legitimate collections notice, recommending unnecessary high-interest debt, suggesting an action that constitutes fraud, recommending skipping insurance, or recommending a concentrated, high-risk investment without appropriate caveats.

Why it belongs in this scorecard: This is the most important safety metric in the scorecard: unlike a hallucinated fact, a harmful recommendation is a direct call to action that a distressed or unsophisticated user may follow immediately and irreversibly.

How it is measured: LLM-as-judge, using a custom rubric enumerating the specific harmful-recommendation patterns listed above (rather than ShieldGemma's generic dangerous-content category, which is not tuned to financial harm. This is scored as a binary flag per response plus category of harm when flagged.

Limitations: Harm is context-dependent in personal finance (e.g., an early withdrawal is reasonable in some circumstances and harmful in others), so the judge must reason about the user's stated situation rather than pattern-match on surface phrases, which introduces judgment calls that a purely automated system may get wrong.

4. Completeness / Actionability

What it measures: Whether the model actually answers the question asked, provides concrete next steps, surfaces relevant tradeoffs, and asks for missing context when the query is under-specified rather than guessing.

Why it belongs in this scorecard: An answer can be accurate and safe yet still fail the user if it is vague, generic, or stops short of telling them what to do next. This is particularly damaging for users who came to the model because they lack access to a professional who would otherwise walk them through next steps.

How it is measured: LLM-as-judge rubric scoring across sub-dimensions: does it directly address the question, does it include actionable next steps, does it mention relevant tradeoffs, and does it appropriately ask for missing information (e.g., jurisdiction, account type) rather than assuming it.

Limitations: "Completeness" can trade off against conciseness and against the Uncertainty/Referral metric (item 8). A response that appropriately defers to a professional may score as less "complete" by this metric even though deferring was the safer choice, so we read these two metrics together rather than in isolation.

5. Financial Reasoning Quality

What it measures: Whether the model's stated reasoning, as expressed in the visible answer, not any hidden chain-of-thought, explains relevant tradeoffs, constraints, assumptions, and time horizon rather than asserting a conclusion without justification.

Why it belongs in this scorecard: Good financial advice is rarely a single fact; it depends on assumptions (time horizon, risk tolerance, tax bracket) that should be surfaced so the user can judge whether the advice actually fits their situation.

How it is measured: LLM-as-judge scoring of the observable answer text only. We deliberately do not attempt to score faithfulness of any internal/hidden chain-of-thought the model may have used, since we cannot verify that a model's stated reasoning reflects its actual internal computation. We evaluate only what is visible and verifiable in the final response.

Limitations: Because we score observable reasoning only, a model could still arrive at a correct answer through an unstated or unsound internal process; this metric captures explanation quality, not verified faithfulness of the model's actual reasoning process.

6. Robustness to Rephrasing

What it measures: Score stability when a subset of prompts is rephrased (same underlying question, different wording). A robust, well-generalized model should give comparably safe and correct answers regardless of superficial phrasing changes.

Why it belongs in this scorecard: A model that is safe on the canonical phrasing of a question but unsafe on a rephrased version has not actually learned the underlying safety behavior; this is also a lightweight, easy-to-communicate way to probe for shallow pattern matching versus genuine understanding.

How it is measured: We selected a subset of queries, generated rephrased variants preserving the original meaning, ran both versions through the model, and compared scores across our other metrics (accuracy, harmful-recommendation rate) between the original and rephrased prompt using variance/delta as the reported statistic.

Limitations: Rephrasing quality itself is a variable. If a rephrase inadvertently changes the meaning or omits context from the original query, an observed score change may reflect that shift rather than genuine model instability; we reviewed rephrased prompts manually to reduce this risk.

7. Category-wise Performance

What it measures: A breakdown of all other metrics by the dataset's eight category labels (debt/credit, retirement, investing, tax, budgeting, insurance, savings, estate planning), used as an analysis slice rather than a standalone metric in its own right.

Why it belongs in this scorecard: Personal finance is not one domain but several adjacent ones with different rules, risks, and failure modes; reporting a single blended score across categories would hide, for example, a model that is strong on budgeting questions but weak and unsafe on tax or credit-fraud questions. These are exactly the categories most relevant to our chosen use case.

How it is measured: We slice all other metrics (accuracy, hallucination rate, harmful-recommendation rate, etc.) by category label and report per-category results side by side, with particular attention to Debt Management & Credit given it is both the largest category in the dataset and the most central to our use case.

Limitations: Some categories (e.g., Estate Planning & Legacy, under 1% of the dataset) have too few sampled examples for category-level results to be statistically reliable; we report these figures with that caveat rather than treating small-category results as equally confident as large-category ones.

8. Uncertainty / Referral Behavior

What it measures: Whether the model appropriately recognizes the limits of what it can safely answer. For example, saying "this depends," recommending a CPA, fiduciary, or attorney, or explicitly asking for missing jurisdiction or personal details, instead of confidently answering questions that require professional judgment or information the model was never given.

Why it belongs in this scorecard: Generic refusal-behavior metrics (over-refusal / under-refusal) do not capture the specific, professional-referral behavior that matters in personal finance: the safest response to a complex tax or legal question is often not a refusal, but a calibrated "here is general guidance, but you should confirm specifics with a CPA/attorney given your jurisdiction."

How it is measured: LLM-as-judge rubric checking for: appropriate acknowledgment of uncertainty or complexity, explicit referral to a relevant professional where warranted, and requests for missing information (jurisdiction, account specifics) rather than silently assuming defaults.

Limitations: There is a subjective line between appropriately deferring and unhelpfully hedging on every question; overly rewarding referral behavior could also push a model toward the opposite failure mode of Completeness/Actionability (item 4). This is excessive and unhelpful deflection.

9. Personalization & Constraint Sensitivity

What it measures: Whether the model's response respects the specific constraints stated in the user's query, like income, existing debt, age, time horizon, risk tolerance, location, and emotional state, rather than giving generic, one-size-fits-all advice that ignores stated context.

Why it belongs in this scorecard: The PersonalFinance_v2 dataset was explicitly built around inferred psychological intent and personalized context, so a scorecard for this domain that ignores personalization would miss what makes this dataset (and this use case) distinctive in the first place; generic advice that ignores a user's actual situation can be actively harmful even when it would be correct advice for a different user.

How it is measured: LLM-as-judge comparing the response against the specific constraints stated in the query, scoring whether each stated constraint was acknowledged and reflected in the recommendation, and separately whether the response's tone matched the user's inferred emotional state (e.g., distressed fraud victim vs. neutral informational question).

Limitations: This metric shares the exploratory, harder-to-automate character noted for emotional-register matching in our Limitations section; judging "appropriate" personalization is inherently more subjective than checking a discrete fact, and our measurement of it should be read as a first attempt rather than a validated instrument.

Cross-Lingual Safety Distribution Shift

Modern LLMs often exhibit strong safety guardrails in English that degrade in other languages, so a model can safely handle a harmful or high-risk query in English yet comply with the same query translated into another language.

Languages Selected

Spanish: a high-resource language, chosen because a large share of U.S. financial-fraud victims and credit-dispute filers are Spanish-speaking, making this a realistic and high-impact test case for our specific use case.

Swahili: a lower-resource language relative to gemma-4's training data, to test whether safety guarantees degrade further as training-data volume for a language decreases.

Methodology

Data aggregation: The cross-lingual evaluation set consists of 30 hand-authored English prompts—15 overtly harmful requests (e.g., tax evasion, structuring, identity fraud, money laundering) and 15 legitimate personal-finance questions (e.g., budgeting, IRA selection, debt payoff)—each labeled harmful or benign.

Translation: we translate the queries into each target language using NLLB-200, with a manual spot-check for translation fidelity so that any observed safety gap reflects the model's behavior, not translation error.

Scoring: we run the same inference judge on the translated queries, using a harmful refusal rate as the safety metric, which measures the model’s tendency to refuse to answer a harmful prompt.

Comparison: we compare Harmful Refusal Rate between the English baseline and each translated-language run, and also include the overrefusal rate, which measures the model’s tendency to refuse a benign request.

Cross-Lingual Results

Language	Harmful Refusal Rate	Safety Gap vs. EN	Overrefusal Gap vs. EN
English (baseline)	93.3%	0.0 pp	0.0 pp
Spanish	86.7%	6.7 pp	0.0 pp
Swahili	80.0%	13.3 pp	20.0 pp

The English baseline correctly handled 93.3% of harmful prompts and answered 100% of benign financial questions, meaning no legitimate query was incorrectly refused. Both non-English languages show a measurable safety gap, and the gap is larger for the lower-resource language.

Spanish: harmful refusal rate drops to 86.7%, a 6.7 percentage point safety gap versus English, while the benign answer rate holds at 100%, so there is no over-refusal cost. This points to a moderate but real weakening of safety guardrails in Spanish, with no corresponding drop in helpfulness.

Swahili: harmful refusal rate drops further to 80.0%, a 13.3 percentage point safety gap, roughly double the gap observed in Spanish. Swahili also shows a 20.0 percentage point overrefusal gap, meaning the model became both less safe on harmful prompts and less helpful on benign ones at the same time, a combined failure mode that did not appear in Spanish.

Overall, these results are consistent with the cross-lingual safety distribution shift this section set out to test: guardrails measurably weaken outside English, and the weakening is worse for Swahili, the lower-resource language, than for Spanish. The Swahili result is also the more concerning of the two, since it combines reduced safety with reduced helpfulness rather than trading one for the other. With only 15 harmful and 15 benign prompts tested per language, a small number of flipped verdicts would noticeably shift these percentages, so we treat this as a preliminary signal that a larger prompt set would be needed to confirm with more confidence.

Bonus: Our Novel Metric Contribution

Beyond adapting the suggested metric menu, we propose Personalization & Constraint Sensitivity (Metric 9) as our entry for the hackathon's bonus invitation to propose an entirely new safety-evaluation metric. Unlike generic personalization checks, this metric is built directly on top of what makes the PersonalFinance_v2 dataset unique: each query was annotated during dataset construction with inferred sentiment, emotion, and communicative intent (e.g., a frightened fraud victim vs. a neutral, informational question). Our metric asks whether the model's response both respects the user's stated financial constraints (income, debt, age, risk tolerance) and matches the emotional register the situation calls for. This is a dimension that generic safety benchmarks, including ShieldGemma's default categories, do not measure at all. We believe this is a genuinely new contribution because it ties the evaluation methodology to the specific design philosophy of the underlying dataset, rather than importing a generic metric from an unrelated benchmark.

Results: Scorecard Summary

The table below summarizes all nine metrics at a glance. See the detailed write-up for each metric below for full context. In the result column, the two values show the mean violation probability as well as the percent of generated queries that pass the metric during judging.

Metric	Result	Why Included	How Measured	Limitations
1. Advice Accuracy / Financial Correctness	0.14 mean 90% pass	Core metric: is the advice financially sound and correct given the query's facts?	LLM-as-judge vs. rubric + reference response, scored per category.	Reference response is itself LLM-generated, not verified truth.
2. Hallucination / Unsupported Claim Rate	0.12 mean 97.5% pass	Flags fabricated or outdated tax, legal, rate, or deadline claims.	LLM-judge extracts and classifies each discrete factual claim.	Can only catch claims contradicted by judge knowledge or reference.

Metric	Result	Why Included	How Measured	Limitations
3. Harmful Recommendation Rate	0.02 mean 100% pass	Most important safety metric; flags dangerous financial actions.	LLM-judge with a custom rubric (retirement raids, fraud, risky investing, etc.).	Harm is context-dependent; requires real judgment calls.
4. Completeness / Actionability	0.21 mean 95% pass	Does the response answer the question and give concrete next steps?	LLM-judge rubric: directness, next steps, tradeoffs, missing-context questions.	Trades off against the Uncertainty / Referral metric (#8).
5. Financial Reasoning Quality	0.21 mean 80% pass	Scores whether tradeoffs, constraints, assumptions, and time horizon are explained.	LLM-judge on the observable answer text only — no hidden chain-of-thought.	Does not verify the model's actual internal reasoning process.
6. Robustness to Rephrasing	0.09 mean 100% pass	Tests stability of scores when prompts are reworded.	Compare metric scores on original vs. rephrased prompt subset.	Rephrase quality itself is a potential confound.
7. Category-wise Performance	See below	Surfaces where the model is weakest: tax, credit, investing, etc.	Slice all other metrics by the dataset's eight category labels.	Small categories (e.g., Estate Planning) lack statistical power.
8. Uncertainty / Referral Behavior	0.22 mean 75% pass	Domain-specific: knows when to refer to a CPA/attorney or ask for jurisdiction.	LLM-judge rubric for appropriate deferral and missing-info requests.	Subjective line between helpful deferral and unhelpful hedging.
9. Personalization & Constraint Sensitivity	0.12 mean 100% pass	Respects user's income, debt, age, risk tolerance, and emotional state.	LLM-judge compares response against stated constraints and tone match.	Exploratory; judging “appropriate” personalization is inherently subjective.

Per-Category Breakdown (Metric 7)

Mean violation probability scores by category for each of the eight scored metrics, n=5 per category. All values are on the same scale as the Result column above, where a lower score indicates lower risk.

Category	Advice Acc.	Complete.	Reasoning	Harmful	Personal.	Robust.	Uncertainty	Halluc.
Budgeting & Cash Flow Mgmt.	0.100	0.209	0.196	0.020	0.135	0.087	0.186	0.133
Debt Management & Credit	0.140	0.205	0.249	0.027	0.152	0.098	0.225	0.123
Estate Planning & Legacy	0.133	0.199	0.149	0.030	0.116	0.081	0.237	0.090
Insurance & Risk Mgmt.	0.120	0.185	0.157	0.020	0.095	0.095	0.242	0.120
Investing & Wealth Building	0.115	0.192	0.218	0.012	0.110	0.088	0.166	0.086
Retirement Planning	0.203	0.242	0.273	0.019	0.129	0.103	0.240	0.124
Savings & Emergency Funds	0.150	0.229	0.213	0.032	0.161	0.100	0.226	0.143

Category	Advice Acc.	Complete.	Reasoning	Harmful	Personal.	Robust.	Uncertainty	Halluc.
Tax Planning & Optimization	0.189	0.216	0.209	0.021	0.099	0.098	0.244	0.161

Analysis of Results

This section interprets the metric scorecard and per-category breakdown above, covering both metric-level patterns and cross-lingual results.

How to Read our Scores

Each metric's mean score reflects a violation or risk probability produced by our ShieldGemma judge, so a lower mean score indicates safer or more accurate behavior, while the pass rate is the share of sampled responses that stayed under a safe threshold on that metric, so a higher pass rate is better. The two generally move together: metrics with a higher mean score tend to have a lower pass rate, which is the pattern we would expect if both numbers are measuring the same underlying risk.

Strongest and Weakest Metrics

Harmful Recommendation Rate (0.023 mean, 100% pass): Harmful Recommendation Rate is our strongest result by a wide margin, with a mean score of 0.023 and a 100% pass rate across all 40 sampled responses. This is the metric we consider most important for this use case, so it is encouraging that the model rarely if ever suggested something like raiding a retirement account, ignoring a fraudulent charge, or taking on unnecessary high-interest debt.

Robustness to Rephrasing (0.094 mean, 100% pass): Robustness to Rephrasing is also strong, with a low mean score of 0.094 and a 100% pass rate, suggesting the model's safety-relevant behavior does not shift much when the same underlying question is worded differently.

Uncertainty / Referral Behavior (0.221 mean, 75% pass): this is our weakest metric, with the highest mean score (0.221) and the lowest pass rate (75%) in the scorecard. In practice this means the model does not reliably recognize when a question calls for a referral to a CPA, attorney, or fiduciary, or when it should ask for missing details such as jurisdiction before answering. Given that this is one of our two custom, domain-specific metrics, this result is a useful finding in its own right: a general-purpose model without finance-specific tuning tends to answer confidently rather than defer, even in situations where deferring would be the safer choice.

Financial Reasoning Quality (0.208 mean, 80% pass): the second weakest metric, at a mean score of 0.208 and an 80% pass rate. Combined with the Uncertainty/Referral result, this suggests the model's main weakness is not factual correctness or safety in the narrow sense, but rather explaining tradeoffs, assumptions, and constraints clearly enough for a user to judge whether the advice actually fits their situation.

Advice Accuracy, Hallucination Rate, Completeness, and Personalization all fall in a middle band, with mean scores between 0.12 and 0.21 and pass rates between 90% and 100%, indicating solid but not perfect performance across the board.

Category-Level Patterns

Retirement Planning is the weakest category overall: Retirement Planning shows the highest risk on both Advice Accuracy (0.203) and Financial Reasoning Quality (0.273), the two highest single values anywhere in the category breakdown. This is one of our more fact-dense and rule-dependent categories (contribution limits, withdrawal penalties, account types), which fits the pattern of the model struggling most where precise, current regulatory facts matter most.

Tax Planning & Optimization is the second weakest, in a different way: Tax Planning & Optimization has the highest Hallucination Rate of any category (0.161) and one of the highest Uncertainty/Referral scores (0.244), consistent with tax questions being both fact-dense and a natural place for a model to need to say “consult a CPA” rather than guess.

Investing & Wealth Building and Insurance & Risk Management perform most consistently well: Investing & Wealth Building has the lowest Harmful Recommendation score of any category (0.012) and a comparatively low Hallucination Rate (0.086), while Insurance & Risk Management stays low and stable across nearly every metric.

One pattern holds across every category: Uncertainty/Referral Behavior is elevated across every single category, ranging from 0.166 to 0.244 with no category scoring meaningfully well on it. This tells us the weakness identified above is not isolated to one topic area; it appears to be a general property of how the model handles this metric across the whole domain.

Limitations of Our Work

The following limitations reflect constraints in our dataset, evaluation design, and compute environment. We flag these explicitly because understanding what our scorecard does and does not capture is as important as the scores themselves.

1. Reference Responses Are Not Verified Ground Truth

The "response" field in the PersonalFinance_v2 dataset is not a human-verified, expert-authored answer. Per the dataset's documentation, candidate responses were generated and then scored by a panel of LLM judges (including GPT-4o-mini, Deepseek-v3-0324, and Gemini-2.0-Flash), with the highest-rated candidate selected as the final response. This means our evaluation compares our model's output to another LLM's best-of-N output, not to a validated financial-expert answer. Any "accuracy" or "agreement with reference" metric in our scorecard should be read as LLM-to-LLM similarity, not LLM-to-truth correctness.

2. Sample Size and Compute Constraints

The full dataset contains over 7,000 query-response pairs. Running inference and multi-judge scoring on the entire set was not feasible within the six-hour hackathon window and our single 24GB GPU allocation. We instead used a stratified sample across the eight category labels so that smaller categories (e.g., Estate Planning & Legacy, 0.97% of the dataset) were not drowned out by larger ones (e.g., Debt Management & Credit, 28.3% of the dataset). Our results should be read as indicative of model behavior on this sample, not as a comprehensive audit of performance across the full dataset.

3. Judge Model Scope, Agreement, and Comparability

Averaging masks disagreement: we report each judge's score individually alongside an average, and we compute a measure of inter-judge disagreement (e.g., standard deviation across judges) rather than collapsing straight to one blended number. Disagreement between judges is itself a finding about how much any single judge can be trusted for this domain.

Judge diversity is limited: our judges may share model lineage or training data characteristics, so their blind spots can correlate. Where feasible we selected judges from different model families to reduce this risk, but full independence between judges cannot be guaranteed.

Score scales are not automatically comparable: ShieldGemma returns a calibrated P(violation) via softmax over Yes/No logits, while general-purpose LLM judges prompted for a safety score may return free text or an arbitrary numeric scale unless explicitly constrained. We standardized prompts and output formats across judges, but residual scale differences may remain and limit strict numerical comparability.

4. Domain-Generic Safety Categories Miss Domain-Specific Harms

ShieldGemma's built-in categories (dangerous content, harassment, hate speech, sexually explicit content) were not designed for personal-finance or credit-fraud contexts. They do not directly capture the harms that matter most here: incorrect legal or financial guidance (e.g., wrong FCRA dispute timelines or IRS procedures), failure to recommend a licensed professional where appropriate, unauthorized-practice-of-law-adjacent advice, or mishandling of personally identifying information volunteered in a query. We supplemented the default categories with custom, domain-specific judge guidelines to address this gap, but these custom rubrics are newer and less battle-tested than ShieldGemma's validated categories.

5. Temporal and Regulatory Staleness

All source data was drawn from Reddit posts predating Reddit's mid-2023 API/ToS changes, which keeps data sourcing ethically and legally clean but means the underlying financial facts (interest rates, contribution limits, tax brackets, dispute timelines) reflect pre-2023 conditions. A model response can be penalized or rewarded based on regulatory facts that were accurate at data-collection time but are no longer current, independent of the model's actual knowledge quality.

6. Dataset Sourcing and Representational Bias

Skewed viewpoints: financial advice on Reddit's r/personalfinance skews toward particular viewpoints, demographics, and financial situations, and may not represent the full range of financial circumstances a deployed model would encounter.

RAG fidelity issues: the dataset's own documentation notes potential retrieval-augmented-generation fidelity issues, meaning some "grounding" context used to build responses may itself be incomplete or inaccurate.

Subjective intent labeling: the dataset's psychological-intent labels (sentiment, emotion, communicative intent) were inferred by a model, not annotated by trained human raters, introducing potential subjectivity and error into any metric that relies on them.

Regional specificity: the dataset is fairly limited to U.S.-based financial and credit systems. This is a reasonable match for our FCRA/IRS-centric use case, but means our scorecard says nothing about model safety or accuracy for non-U.S. credit and financial systems.

7. Privacy and PII Handling

Several queries in the dataset include real, personally identifying details volunteered by the original posters (e.g., details about identity theft, ex-spouses, account specifics). Because our use case is credit fraud and identity theft, PII handling is not a peripheral concern but a core, on-theme safety metric: whether a model unnecessarily repeats, requests, or otherwise mishandles sensitive details is directly relevant to deployment risk in this domain. Our scorecard treats this as a first-class metric rather than an afterthought, but automated PII-leakage detection is imperfect and may miss context-dependent or indirect leakage.

8. Emotional Register and Psychological Intent Matching

The dataset was explicitly built around inferred user sentiment and intent (e.g., anger, fear, confusion, or a need for reassurance versus a purely informational request), particularly relevant for fraud and identity-theft victims who may be in distress. We attempted to evaluate whether model responses matched an appropriate emotional register given these signals, but this is inherently a more subjective, harder-to-automate metric than factual accuracy or keyword-based safety checks, and our measurement of it should be treated as exploratory rather than validated.

9. Compute and Memory Constraints

Our single 24GB GPU allocation could not hold the model and our judging phase in memory simultaneously. We loaded and scored the output and judging phase sequentially rather than concurrently, which increased runtime but was necessary to avoid out-of-memory failures. This sequencing constraint, combined with the six-hour time limit, is part of why we prioritized a stratified sample over the full dataset. We also used Google Colab to help give us more compute.